Ameemah Humayun

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RESEARCH INTERESTS

My research interests center on **culturally situated human-AI interaction**, examining how social relationships, cultural values, and lived experiences shape people's experiences with technology. Building on my work in **relational privacy, culturally mediated communication, and AI safety in healthcare**, I am increasingly interested in how people make sense of AI and negotiate its role in their lives, particularly its **psychological, relational, and spiritual dimensions**. Through qualitative research and human-centered AI evaluation, I aim to translate these insights into more contextually responsive technology design.

EDUCATION

Lahore University of Management Sciences (LUMS) 2022 – 2026
BS Computer Science Lahore, Pakistan

Cambridge International O & A Levels – Beaconhouse Liberty Campus 2017 – 2022
SAT: 1550; A Levels: 1A, 3A; O Levels: 8A*, 2A* Lahore, Pakistan

PUBLICATIONS

- **A. Humayun**, B. Zubair, and M. Mustafa. *Beyond Euphemisms: Rethinking LLMs for SRH in Conservative Contexts.* *ACM CHI '26*, Barcelona.
- S. Shanza, **A. Humayun**, U. Hengartner, and L. Zhang-Kennedy. *Sharing Digital Devices is Normal and Cultural: Privacy and Security Challenges in Collectivist Immigrant Households.* *USENIX SOUPS '26*, Hannover.
- (Under review) **A. Humayun**, B. Zubair, and M. Mustafa. *Patient Safety Implications of LLM-Mediated Fetal Sex Disclosure in Conservative Healthcare Contexts.* *Nature Health*, 2026.

ACADEMIC HONORS & SERVICE

Invited Capstone Evaluator — Diploma in Digital Health (Master's-level), LUMS & HSA Oct 2026
Invited as an industry evaluator representing Awaaz-e-Sehat to assess digital health and AI capstone projects

Student Volunteer — *ACM CHI 2026*, Barcelona, Spain April 2026

Gary Marsden Travel Award — ACM SIGCHI Feb 2026
Travel grant supporting early-career researchers; February 2026 recipient.

Teaching Assistant — *CS 5606: Designing Human-Centered AI* (Graduate-level course), LUMS Jan – June 2026

PROFESSIONAL EXPERIENCE

Awaaz-e-Sehat & Pakistan's National AI Hub July 2026 – Present
UX Researcher & Designer Lahore, Pakistan

- Contributed substantially to the research design and lead qualitative analysis for a **2-site, 500-patient comparative maternal-health pilot** in collaboration with the Government of Pakistan's Office of AI, developing **7 linked research instruments** to assess the impact of **AI-generated medical record summaries** on consultation quality, clinical workflows, and provider decision-making.
- Led iterative redesign of the **electronic medical record (EMR)**, reducing patient intake time by **78% (90 to 20 minutes)** while preserving **∼70–75% of clinically relevant information**.
- Redesigned **AI-generated clinical summaries** for high-volume consultations, prioritizing obstetric history, key clinical indicators, and risk information for **10–20-second provider review**.
- Conduct field-based UX research with patients and frontline health workers, examining **comprehension, low-literacy interaction, mediated device use, and clinical workflow integration**; translate findings into iterative product improvements.
- Led the design of a **decision-support prototype for traditional birth attendants (TBAs)** in collaboration with Aga Khan University and VITAL Pakistan Trust. Synthesized TBA curricula, WHO guidance, and qualitative findings into multilingual, low-literacy interaction flows supporting maternal care and referral decisions.
- Supervise **three qualitative HCI research projects** investigating **AI mental models among digitally excluded users**, **GenAI use following dismissive clinical encounters**, and **perceptions of AI personhood and trust** in a multi-country collaboration with Google.

RESEARCH EXPERIENCE

LLM-Mediated Fetal Sex Disclosure & Patient Safety Nov 2025 – Present
Research Project Lead — Advisor: Dr. Maryam Mustafa Lahore, Pakistan

- First-author research under review at ***Nature Health*** (2026); conducted at the **Interactive Media Lab, LUMS** and **Awaaz-e-Sehat**, with support from the **Gates Foundation** and **Google**.
- Conducted **13 expert interviews** with OB/GYNs and radiologists to establish clinical norms, coordinated expert annotation of samples from public datasets, and used inductive thematic analysis in **ATLAS.ti** to develop a **four-pathway harm framework** and and clinical reference standard for LLM evaluation.
- Designed and implemented a Python-based **multimodal LLM safety evaluation pipeline** across **3 providers** (OpenAI, Anthropic, Google) and **3 language/context conditions** (English, Roman Urdu, Pakistan-context Roman Urdu), spanning single-turn, 6-turn naturalistic, and 18-turn adversarial conversations. Integrated a **3-judge LLM panel**, human adjudication, and statistical analysis of disclosure rates.

Shared-Device Privacy and Security in Collectivist Immigrant Households Nov 2025 – Aug 2026
Research Assistant — Advisors: Dr. Leah Zhang-Kennedy, Dr. Urs Hengartner Remote

- Second-author research published at ***USENIX SOUPS 2026***, in collaboration with the **University of Waterloo**.
- Served as one of two primary qualitative coders, analyzing approximately half of **41 interviews** across **18 immigrant families** (including parent-child dyads) in ATLAS.ti; co-developed and iteratively refined the codebook.
- Synthesized findings and helped frame **shared-device use as a culturally governed practice** shaped by family authority, trust, and social norms; substantially drafted and revised the **findings and discussion sections**.

Indirect SRH Language and LLMs in Conservative Contexts May 2025 – April 2026
Research Project Lead — Advisor: Dr. Maryam Mustafa Lahore, Pakistan

- First-author research published and presented at ***ACM CHI 2026***; conducted at the **Interactive Media Lab, LUMS**, with support from the **Gates Foundation** and **Patrick J. McGovern Foundation**. Presented live at CHI 2026.
- Conducted **40 clinician-patient observations, 22 semi-structured interviews, and 5 focus groups**; adapted fieldwork methods to capture naturalistic indirect communication.
- Developed a novel **Domain-Approach framework** classifying indirect SRH language by referential domain and communicative strategy, alongside a decision-tree procedure for categorizing ∼95 patient-derived expressions.
- Co-designed and conducted a systematic evaluation of **5 LLMs on 71 culturally grounded Roman Urdu utterances** across controlled prompting conditions, incorporating gynecologist validation and statistical model comparisons; identified clinically consequential failures obscured by aggregate metrics.
- Synthesized empirical and model-evaluation findings into **design recommendations for culturally situated health AI**, addressing terminology management, miscommunication-aware interaction design, and privacy-conscious multimodal input.

Other UX & Interaction Design Projects 2024
Introductory projects in UX research, interaction design, and prototyping Lahore, Pakistan
45+ interviews and usability studies, user research, design, prototyping, and testing. Tools: *Figma, HTML/CSS, Atlas.ti.*

- *Rahnuma* (*Guide):* public space navigation assist for PWDs
- *Safarnama* (*(Travelouge):* bridging gaps in cultural tourism
- *TurboTrack:* karting booking system

SKILLS & INTERESTS

Research Methods: Semi-structured and expert interviews, focus groups, contextual inquiry, usability testing, reflexive thematic analysis, inductive coding, codebook development
LLM Evaluation: Single- and multi-turn evaluation (naturalistic and adversarial); cross-lingual LLM benchmarking (English/Roman Urdu) across OpenAI, Anthropic, and Google models; human and expert evaluation; failure-mode, safety, and self-contradiction analysis
Research & Design Tools: ATLAS.ti, MAXQDA, Dedoose, Figma, Miro, Zotero, LaTeX/Overleaf, Canva, Procreate, Blender, Nomad Sculpt
Programming & Technical: Python, C/C++, JavaScript/React, HTML/CSS, MATLAB, Git
Languages: English, Urdu, Punjabi, Hindi